



## Venn Diagrams & Geometric Representa

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SUBJECT

Mathematics Foundation T

SEMESTER

Semester 1

### // Notes Summary

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{ }
title: "Venn Diagrams Geometric Representation"
semester: subject: "Mathematics Foundation to
Computer Science I" subjectSlug: "mathfoundation1"
unit: "Unit I: Set, Relation and Function" unitSlug:
"unit1setrelationfunction" order: description:
"Visual set representations, geometric regions for
```

# NOTES THAT STICK.

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Topics Covered

Intermediate

Difficulty Level



## Why this topic exists

Venn diagrams serve as the primary geometric representation for set theory operations ( $\cup, \cap, \setminus, '$ ). Rather than evaluating membership or proving set identities purely through symbolic logic, Venn diagrams map sets directly to closed spatial regions. In computer science, this visual model underpins relational database JOIN operations (SQL), Boolean query construction in information retrieval systems, and probability theory for compound events.

## What is a Venn Diagram?

### ✦ [DEFINITION] Venn Diagram Structure

A **Venn diagram** represents sets as closed geometric regions (typically circles) contained within a bounding rectangle. - **Universal Set ( $U$ )**: Represented by the outer bounding rectangle. - **Sets ( $A, B, \dots$ )**: Represented by closed circles inside the rectangle. - **Regions**: Overlapping areas denote shared elements; non-overlapping areas denote elements unique to a set.

## Representing Basic Set Operations

The four core set operations correspond to specific spatial regions in a diagram:

- **Union ( $A \cup B$ )**: The entire region bounded by both circles  $A$  and  $B$  combined. Represents elements in  $A$ ,  $B$ , or both.
- **Intersection ( $A \cap B$ )**: The overlapping central region common to both circles. Represents elements shared by both sets.
- **Difference ( $A - B$  or  $A \setminus B$ )**: The crescent-shaped region of circle  $A$  excluding any area shared with  $B$ . Represents elements strictly in  $A$ .
- **Complement ( $A'$  or  $A^c$ )**: The entire area within rectangle  $U$  strictly outside circle  $A$ . Represents all elements in the universal set that do not belong to  $A$ .

## Set Cardinality with Venn Diagrams



For any two finite sets  $A$  and  $B$ :

✓ [VALID / RULE] Principle of Inclusion-Exclusion (2 Sets)

$$|A \cup B| = |A| + |B| - |A \cap B|$$

If  $A$  and  $B$  are disjoint ( $A \cap B = \emptyset$ ):

$$|A \cup B| = |A| + |B|$$

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